acceleration, according to Galileo, is independent of its nature and its velocity. The earth, of course, must be decisive for the existence of this acceleration. It seemed, then, that the earth by its mere presence exerted a force upon the body. The earth consists of many parts. The idea seemed inevitable that each of these parts affects the falling body and that all these effects are combined. There seems then to be a force which bodies by their very presence exert upon each other through space. These forces seem to be independent of velocities, dependent only upon the relative position and quantitative property of the various bodies exerting them. This quantitative property might be conditioned by its mass, for the mass seems to characterize the body from the mechanical point of view. This strange effect of things at a distance may be called gravitation.

Now to gain precise knowledge of this effect, one has only to find out how strong is the force exerted upon each other by two bodies of given mass from a given distance. As for their direction, it would probably be no other than the line connecting them. Finally then, what remains unknown is only the dependence of this force upon the distance between the two bodies. But this one cannot know *a priori*. Here, only experience could be of use.

Such experience, however, was available to Newton. The acceleration of the moon was known from its orbit and could be compared with the acceleration of the freely falling body on the surface of the earth. Furthermore, the movements of the planets about the sun had been determined by Kepler with great exactness and comprehended in simple empirical laws. So it was possible to ascertain how the effects of gravitation coming from the earth and those coming from the sun depended on the factor of distance. Newton found that everything was explainable by a force which was inversely proportional to the square of the distance. And with that the goal was reached, the science of celestial mechanics was born, confirmed a thousand times over by Newton himself and those who came after him. But how about the rest of physics? Gravitation and the law of motion could not explain everything. What determined the equilibrium of the parts of a solid body? How was light to be explained, how electrical phenomena? By introducing material points and forces of various kinds acting at a distance, everything seemed in a fair way to be derivable from the law of motion.

That hope has not been fulfilled, and no one any longer believes in the solution of all our problems on this basis. Nevertheless, the thinking of physicists today is conditioned to a high degree by Newton’s fundamental conceptions. It has so far not been possible to substitute for the Newtonian unified conception of the universe a similarly unified comprehensive conception. But what we have gained up till now would have been impossible without Newton’s clear system.

From observation of the stars have chiefly come the intellectual tools indispensable to the development of modern technique. For the abuse of the latter in our time creative intellects like Newton’s are as little responsible as the stars themselves, contemplating which their thoughts took wing. It is necessary to say this, because in our time esteem for intellectual values for their own sake is no longer so lively as it was in the centuries of the intellectual renascence.